OSTREA MINERAL LABORATORIES, INC.
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July 02, 2018

Ref №: Q-1806-061 Rev.2

PRICON MICROELECTRONICS, INC.
No.14 Ampere St. Brgy. Diezmo, LISP-I
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Tel # (02) 843-0870
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ATTENTION: MR. JOSHUA L. SARMIENTO

CC: MS. REBECCA D. VALENZUELA

1 SUBJECT: WATER / SOURCE / AMBIENT / WORK ENVIRONMENT MEASUREMENT

Indoor Air Quality Monitoring is a process composed of interrelated activities which include sampling, analysis, data interpretations and reporting. It is an essential tool that will help determine what job assignments and work stations are potential sources of pollution, to identify predominant sources of specific types of air contaminants and to determine compliance with DENR Administrative Order 2000-81 and Occupational Safety and Health Standards of DOLE. The results will be the basis for setting up control programs to eliminate or reduce pollution and hazard

2 SCOPE OF WORK

The sampling and analysis will be conducted as follows:

2.1 SOURCE

Type of Sampling	Type of Source	№ of Source	Parameter
Source Emission	Genset 1– 750 KVA Caterpillar Standy Genset	1	SOx, NOx, CO (3x trials)

2.2 AMBIENT

Type of Sampling	№ of Stations	Parameter
AMBIENT (1 hour monitoring)	5 stations (Gate 1 & 2, Near Chemical Storage Area, Near Kitchen (material entry) Area & Front of unloading bay)	TSP, SO_2, NO_2, Pb, CO, Noise

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2.3 Work Environment Measurement

Physical Parameters					
Sampling Stations	WBGT (Heat Stress)	%RH	Ventilation Rate	Noise	Dust Particle (TND)
TS	5	5	10	5	2
CN	10	20	10	10	4
YF	5	5	6	4	4
PPS	7	6	7	10	2
Support	6	6	7	4	0
Total	**33**	**42**	**40**	**33**	**12**

Chemical Parameters					
Sampling Stations	VOC	Lead	Toluene	Ethyl Acetate	IPA
TS	1	1	0	1	1
CN	1	1	0	0	4
YF	1	0	2	1	1
PPS	1	0	0	1	1
Support	0	0	0	0	1
Total	**4**	**2**	**2**	**3**	**8**

3 METHODOLOGIES

3.1 Source

Parameters	Sampling / Analytical Methodology
Sulfur Dioxide (SO_2)	Isokinetic-Barium Thorin Titration - USEPA Method 6 (5/6)
Nitrogen Oxide (as NO_2)	Phenol Disulfonic (PDS) Analysis - USEPA Method 7
Carbon Monoxide (CO)	Carbon Monoxide (NDIR) - USEPA Method 10

3.2 Ambient

Parameters	Methodology
Total Suspended Particulate (TSP)	High Volume Sampler/ Gravimetric (37mm)
Sulfur Dioxide (SO_2)	Gas-Bubbler – Pararosaniline
Nitrogen Dioxide (NO_2)	Gas-Bubbler – Greiss Saltzman
Lead (Pb)	High Volume Sampler - AAS/ICP
Carbon Monoxide (CO)	Direct Reading – CO analyzer
Noise	Sound Level Meter

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3.3 WEM

PARAMETER	METHODOLOGY
Physical Parameters	
Relative Humidity	Direct reading using Psychrometer
Noise	Direct Reading using Sound Level Meter
Air Velocity	
Ventilation Rate	Ventilation rate will be computed based on CO_2
Dust Particle (Total Nuisance Dust-TND)	Gravimetric / NIOSH 0500
WBGT (Heat Stress)	Direct reading using Luxmeter
Chemical Parameters	
VOC	Multigas Analyzer using Photo ionization Detector
Toluene	NIOSH 1501
Ethyl Acetate	NIOSH 1457
IPA	NIOSH 1400
Lead	NIOSH 7082

4 REPORTING

OMLI will submit a written report, with the result of analysis versus standard limit values set by DENR and the Occupational Safety and Health Standards of DOLE. The report will include evaluation of each parameter and summary of findings and results and discussion. Also included are the list of tables of methodologies, observed stack gas parameters and calculated isokinetic variations, measured pollutants concentrations at the exhaust stack, laboratory test certificates and other required documents based on DENR Standard reporting.

5 SERVICE COSTS

Parameter	Qty	Unit Cost (Php)	Total Cost (Php)
Source	1	18,000.00	18,000.00
Ambient	5	6,000.00	30,000.00
WEM:			
Heat Stress	33	110.00	3,630.00
Relative Humidity	42	110.00	4,620.00
Ventilation Rate	40	600.00	24,000.00
Noise	33	110.00	3,630.00
Total Nuisance Dust (TND)	12	670.00	8,040.00
VOC	4	2,600.00	10,400.00
Lead	2	790.00	1,580.00
Toluene	2	6,500.00	13,000.00
Ethyl Acetate	3	6,500.00	19,500.00
IPA	8	5,500.00	44,000.00
Sampling Fee			10,000.00
Subtotal			**190,400.00**

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Note: Company is VAT exempted.
*Once sampling done on weekend or holiday, additional 20K per day
*In compliance with the new DENR Memorandum Circular No. 2007-003, three (3) trial runs are required per parameter per source.
* Quoted price is for reference only and invoice shall depend on the actual work done.
*In case a scheduled sampling is postponed/cancelled AFTER test plan has been approved by the DENR and/or due to unavailability of the facility to be tested and any other reasons within clients control or after OMLI's personnel have been mobilized the client shall be charged **50% of the total project cost**.
*Sampling shall be done on the agreed date from 9:00am to 6:00pm only. Any sampling activities from 7:00pm onwards shall be billed at an additional rate of Php 2,500.00 per hour. Additional Twenty Thousand Pesos per day (Php 20,000.00/day) shall be charged if sampling activities will exceed the allotted sampling dates.

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6 TERMS AND CONDITIONS

- The customer shall provide an on-site single phase 220VAC/60Hz power.
- The customer shall provide scaffolding during sampling when needed.
- The customer shall ensure that fuel-burning equipment should be operating continually at normal to maximum operating load during activity.
- The customer shall provide the necessary operational information during the time of sampling.
- It shall be assumed that client guarantees a smooth flow of operation and any deviation from the normal condition shall be addressed immediately by the client.
- As per USEPA Method 1, stack diameter should be at least ≥30 cm.
- As per USEPA Method 1, stack having a diameter of :
  30 - 150 cm - customer shall provide at least 2 sampling portholes.
  151 - 200 cm - customer shall provide 4 sampling portholes.
- As per USEPA Method 1, sampling portholes should be at least 4 inches/10 cm diameter.
- As per USEPA Method 1, sampling portholes shall be located at least two times (2D) the stack diameter downstream and half of (0.5D) the stack diameter upstream referred from any disturbance.
- For metals sampling, customer shall provide clips and eyebolts for monorail installation.
- Absolute value of the rotation angle should not be >20 degrees. Verification of which will be done during actual sampling.
- Please note that prior to commission of the project, our team shall examine each stack to determine conformance to required porthole configuration and shall give recommendations to the proper configuration.
- For source monitoring, DENR regulations stipulation that monitoring shall commence **30 days** upon submission of sampling plan.

Should you find this proposal satisfactory and acceptable, kindly affix your signature on the space provided below and sends a copy for our file or please provides Purchase Order or its equivalent prior to commencement of the project.

Thank you.

Confirmed:

ANTHONY BABOR
Sales & Marketing Associates

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